# linnest

*a typst interface to the `linnet` crate*

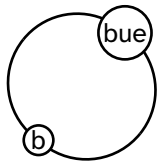
`linnet` is a rust crate that provides a graph data structure and many graph algorithms, and crucially for this package, multiple layout algorithms, as well as a dot parser (using the `dot_parser` crate).

## Contents



## Linnet

The `linnet` crate that this package wraps around, is built around the half-edge graph data structure. This means that instead of a graph being represented as a set of nodes and edges, it is represented as a set of half-edges $H$, and a set of vertices $V$. The graph structure is then encoded through two maps. The first map, $\partial : H \longrightarrow V$ maps each half-edge to its corresponding vertex. The preimage of any vertex $v$ is the set of half-edges that map to it, called the crown of $v$. The second map, $\iota : H \longrightarrow H$, is an involution that *glues* half edges together to form edges. If a half-edge is glued to itself, we call that an external half-edge.




## Linnest Typst APIs

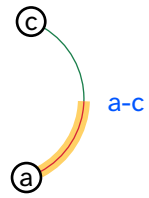
Linnest exposes the `linnest.wasm` graph layout plugin through a small Typst API. Its drawing layer imports the sibling Kurvst package; the standalone Kurvst manual documents the full curve and path-pattern API. Derived edge layers use Kurvst's CeTZ-compatible path wire format internally, so path data can move through Linnest, Kurvst, and CeTZ-style tooling without an extra shape. The public surface is intentionally narrow:

- `graph` for construction, parsing, inspection, joins, and graph algorithms.
- `subgraph` for subgraph object construction and inspection.
- `layout` for the separate layout pass.
- `draw` for rendering a laid-out graph object with CeTZ.
- `physics` for reusable particle-line edge styles.

### Minimal Build Example

```
#import "../src/lib.typ": draw, edge, graph, layout, node, sink, source, subgraph
```

```
#let g = graph.build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>, compass: "e"),
    <a-c>,
    sink(<c>, compass: "w"),
    label: "a-c",
    statements: (
      color: "0055ff",
      source-color: "d72638",
      sink-color: "1b7f4c",
    ),
  )
},
  name: "demo",
)
#let g = layout(g)
#let east = subgraph.compass(g, "e")
#let edges = graph.edges(g, subgraph: east)
#let dot = graph.dot(g)
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.label]
#let source-style(edge) = (stroke: rgb("#" + edge.source-color) + 0.5pt)
#let sink-style(edge) = (stroke: rgb("#" + edge.sink-color) + 0.5pt)
#draw(g, subgraph: east, edge-label: edge-label, source-style: source-style, sink-style: sink-
style)
```




## Graph Objects

Graph and subgraph objects are opaque zero-copy values. Build or parse graph objects with `graph`, transform graph objects with `layout`, and pass objects back to `graph` or `subgraph` for inspection.

- `graph.parse(input)` parses one or more DOT digraphs and returns an array of graph objects.
- `graph.build(..)` constructs one graph object from a stream of node and edge items.
- `node(..)` returns a node item.
- `source(..)` and `sink(..)` return half-edge endpoints.
- `edge(..)` returns an edge item built from source/sink endpoints and an optional edge name.
- `layout(graph, ..)` runs layout as an explicit second step. Its settings are named parameters so calls stay descriptive and Tidy can document each field.
- `draw(graph, ..)` draws a laid-out graph object with CeTZ.
- `graph.dot(graph)` returns a DOT string for inspection or export.

## Graph Specs

The Typst construction API is half-edge first: create node items, create source and sink half-edge endpoints, then pass edge items to `graph.build`. `graph.build` accepts both comma-separated items and ordinary Typst code blocks. Typst labels such as `<a>`, `<h1>`, and `<e1>` are API names; they are resolved before the wire format is sent to the Rust plugin. Numeric `id` arguments are order/index overrides. The `label` argument is display metadata and becomes the DOT `label` statement for both nodes and edges.

```
#let g = graph.build({
  node(<a>)
  node(<c>)
  edge(source(<a>), <e1>, sink(<c>))
})
```

`source(..)` and `sink(..)` accept a node reference plus optional `name`, `id`, `statement`, and `compass`. `name` is a Typst label name for the half-edge; `id` is numeric:

```
edge(source(<a>, name: <h1>, id: 0), <e1>, sink(<c>, id: 2), label: "a-c")
```

One half-edge creates an external edge. The side is determined by the constructor, so there is no public `flow` argument:

```
edge(<incoming>, sink(<a>))
edge(source(<c>), <outgoing>)
```

String-valued DOT statements may contain `{name}` placeholders. `graph.build` expands each placeholder while constructing the graph object. Use `{{` and `}}` for literal braces. Unknown placeholders are left unchanged.

This is useful for graph-level `edge-statements`: the default edge statement is merged into each edge, then expanded against that edge's complete statement dictionary. The following default metadata records a display label derived from the per-edge `label` statement:

```
#let g = graph.build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label: "a-c",
    statements: (color: "0055ff"),
  )
},
  edge-statements: (
    color: "000000",
    display-label: "{label}",
  ),
)
```

## Placements

`graph.pos` creates a first-class placement. The default `mode: "pin"` turns a coordinate into a fixed layout constraint and a drawable position. Use `mode: "start"` when the coordinate should only seed the layout:

```
#let g = graph.build({
  node(<a>, pos: graph.pos(x: -2, y: 0))
  node(<c>, pos: graph.pos(ref: <a>, dx: 4, dy: 0))
  edge(source(<a>), <a-c>, sink(<c>), pos: graph.pos(x: 0, y: 1.2))
})
```

`graph.group` links one coordinate across several nodes or edge control points. A `side` of `"+"` keeps the coordinate positive, and `"-"` keeps it negative. GammaLoop external-edge columns use this to keep incoming and outgoing external legs on opposite sides while pairing rows by a shared `y` group:

```
#let edges = (
  edge(
    source(<right-ext>),
    sink(<center>),
    pos: graph.pos(x: graph.group("right", side: "+"), y: graph.group("edgee0")),
  ),
  edge(
    source(<center>),
    sink(<left-ext>),
    pos: graph.pos(x: graph.group("left", side: "-"), y: graph.group("edgee0")),
  ),
)
```

Raw DOT input uses the same placement model through the `pos` attribute. The standard Graphviz subset is preserved: `pos="x,y"` is a starting coordinate and `pos="x,y!"` is pinned. Linnest extends the value with

explicit id references and axis entries. In axis entries, `!` belongs to that axis: numeric entries without `!` are starting coordinates, numeric entries with `!` are fixed constraints, and grouped entries must use `!` because groups are constraints.

```
digraph {
  a [id=0 pos="0,0!"]
  b [id=1 pos="ref(node:0)+4,0!"]
  a -> b [id=0 pos="ref(node:1)+0,1!"]
  b -> c [id=1 pos="ref(edge:0)+1,0!"]
  c [id=2 pos="x:2!"]
  d [id=3 pos="x:2!,y:1"]
  ext -> a [id=2 pos="x:@-left!,y:@edge0!"]
}
```

The DOT grammar is intentionally id-based: `ref(node:0)` uses a node `id` and `ref(edge:0)` uses an edge `id`. Bare names and implicit edge order are not placement references. The `@` syntax denotes grouped coordinate constraints; `@+name` and `@-name` keep the grouped coordinate on the positive or negative side respectively.

```
pos="x,y"                   // start x and y
pos="x,y!"                  // pin x and y
pos="x:<coord>"             // start numeric x
pos="x:<coord>!"            // pin x
pos="y:<coord>!"            // pin y
pos="x:<coord>!,y:<coord>"  // pin x, start numeric y
pos="x:<coord>,y:<coord>!"  // start numeric x, pin y
```

Draw styling is Typst-native. Pass dictionaries or callbacks to `draw`; edge callbacks receive the merged `scope`, edge statements, source/sink half-edge records, and the edge index:

```
#let edge-label(edge) = text(fill: rgb("#" + edge.color))[#edge.display-label]
#let source-style(edge) = (stroke: red + 0.5pt)
#let sink-style(edge) = (stroke: blue + 0.5pt)
#draw(layout(g), edge-label: edge-label, source-style: source-style, sink-style: sink-style)
```

Marks can follow graph orientation as a first-class draw option. Put the same mark layer on both halves and set `mark-orientation: "edge"`; default-oriented edges mark the source half, reversed edges mark the sink half with the marker flipped, and undirected edges suppress the mark.

```
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: (end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto), scale: 0.75),
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), source-style: oriented-arrow, sink-style: oriented-arrow)
```

## Physics Edge Styles

`physics.style(..)` returns `source-style`, `sink-style`, and `edge-label` callbacks for `draw`. The default source/sink strokes use darker/lighter halves to encode the graph source/sink split. Fermion map entries marked with `fermion-flow` receive one particle-flow arrow on the main edge using `mark-orientation: "edge"`; paired edges follow `edge.orientation`, dangling half edges place the arrow in the middle of the visible half edge, and undirected edges omit the arrow. This is independent of momentum arrows.

Set `momentum-arrows: true` to draw centered parallel arrow decorations while the main edge remains connected to the nodes. The arrows flow from source to sink independently of the DOT `dir`/`orientation` value; there is one momentum marker per edge, even though the base edge is internally styled as source and sink halves. On paired edges the marker lives on the sink half; on dangling edges it lives on the existing source or sink half-edge. The decoration defaults to a plain black `0.55pt` stroke and a scaled CeTZ `"barbed"` arrowhead, so it does not inherit the base particle stroke. The arrow length is capped by both `momentum-arrow-length` and `momentum-arrow-ratio`, so the shorter one wins. When the layout provides an edge-label position, the arrow offset is signed so the momentum marker is drawn on the same side of the edge as that label. Use `momentum-arrow-offset` to set the normal displacement. `momentum-arrow-mark: auto`

uses the default barbed marker, `momentum-arrow-mark: none` draws only the momentum line, and any other value overrides the CeTZ mark style, for example `"stealth"` or `(end: "barbed", scale: 1.6)`. Only the `end` mark is kept so source/sink splitting cannot create a second momentum head.

Optional labels can be built from edge metadata with `show-edge-index`, `show-half-edge-index`, `show-particle`, and, for explicit momentum fields, `show-momentum`. `show-half-edge-index` only emits a value for dangling half edges.

```
#import "../src/lib.typ": draw, edge, graph, layout, node, physics, sink, source

#let g = graph.build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <fermion-main>,
    sink(<c>),
    statements: (particle: "fermion", id: "7"),
  )
  edge(
    source(<c>),
    <fermion-out>,
    statements: (particle: "fermion", id: "8"),
  )
},
  name: "physics",
)
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```

Set `edge-offset` on `draw`, or `offset` in a `source-style`/`sink-style` dictionary, to draw a fitted parallel path. Style callbacks may also return an array of dictionaries; the layers are drawn in order on the same graph edge, so parallel strokes do not require duplicate graph edges.

```
#let base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
)
#let offset-style(edge) = (
  offset: 0.18,
  length: 1.4,
  ratio: 0.5,
  resolve-length: "min",
  stroke: (paint: rgb("#2f6f4e"), thickness: 1pt, cap: "round"),
)
#let source-style(edge) = (base-style(edge), offset-style(edge))
#let sink-style(edge) = (base-style(edge), offset-style(edge) + (mark: (end: ">")))
```

The parallel path is applied to the base edge geometry before patterns and other decorations; node outsets then trim the shifted path, so shifted paths still start and end outside fitted node circles. Add `edge-length` or `length` to center-trim the shifted path to a fixed arc length, and add `edge-ratio` or `ratio` to cap it by a fraction of the base edge length. `edge-resolve-length`
**`resolve-length` decides how to combine both limits** `"min"`/`"shorter"` (default), `"max"`/`"longer"`,

`"length"`/`"fixed"`, `"ratio"`/`"relative"`, `"none"`/`"full"`, or a function receiving `(base-length, length, ratio)`. Set `offset-side: "label"` on an offset layer to choose the sign of `offset` so the layer is drawn on the same side of the curve as the edge label.

`source-style-eval`, `sink-style-eval`, and `label-eval` are ordinary kebab-case edge metadata for downstream renderers or DOT export. They can be set as direct arguments on `graph.build` or `edge`, instead of being manually nested under `edge-statements` or per-edge `statements`:

```
#let g = graph.build({
  node(<a>)
  node(<c>)
  edge(
    source(<a>),
    <a-c>,
    sink(<c>),
    label-eval: "(text(fill: rgb(\"#{color}\"))[{label}])",
    statements: (color: "0055ff", label: "a-c"),
  )
},
  name: "demo",
  source-style-eval: "(stroke: red + 0.5pt)",
  sink-style-eval: "(stroke: blue + 0.5pt)",
)
```

`graph.build` also accepts comma-separated items:

```
#let g = graph.build(
  node(<a>),
  node(<b>),
  edge(
    source(<a>, compass: "e"),
    <ab>,
    sink(<b>, compass: "w"),
    label: "ab",
    statements: (color: "0055ff"),
  ),
  name: "demo",
  statements: (full_num: "x + y"),
  node-statements: (shape: "circle"),
  edge-statements: (
    color: "000000",
    display-label: "{label}",
  ),
)
```

## Graph Queries

`graph.info(g)` returns graph metadata. `graph.nodes(g)` returns node records, and `graph.edges(g)` returns edge records. Pass `subgraph: sg` to filter nodes or edges by an subgraph object.

`graph.join(left, right, key: "statement")` joins matching dangling half edges. The key is read from half-edge data and can be `"statement"`, `"compass"`, or `"id"`.

`graph.cycles(g)` returns subgraph objects for a cycle basis. `graph.forests(g)` returns subgraph objects for spanning forests.

## Layout Model

`layout` starts from a traversal-tree placement, then optimizes the positions of graph nodes and edge control points. The initial tree spacing is

$$L = \lambda \sqrt{\frac{WH}{\max(n, 1)}}$$

,

with horizontal spacing $\tau_x L$ and vertical spacing $\tau_y L$. Here $\lambda$ is `length-scale`, $W$ is `viewport-w`, $H$ is `viewport-h`, $\tau_x$ is `tree-dx`, and $\tau_y$ is `tree-dy`. These fields set the geometry scale for both layout modes.

For deterministic, non-iterative placement, use `layout-algo: "tree"` or `layout-algo: "dot"`. `"tree"` places a traversal forest by levels. `"dot"` uses a directed layered placement for acyclic inputs, falling back to tree placement when the selected edges contain a cycle. Both modes also work on a subgraph:

```
#let g = graph.parse("digraph partial { a -> b; b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree", subgraph: tree)
#draw(g)
```

Only nodes touched by the selected subgraph are placed. Edge control points are then resolved from the current node positions, so edges outside the selected subgraph are drawn as straight lines unless their own position constraints say otherwise.

`layout-algo: "force"` and `layout-algo: "anneal"` also accept `subgraph`. For these iterative modes, nodes and edge control points outside the selected subgraph stay fixed and act as boundary points while the selected subgraph is optimized.

Set `layout-nodes: "fixed"` to keep every node at its current `pos` for this layout pass and move only edge control points. With `subgraph`, only edges in the selected subgraph are moved; all other edge control points keep their current positions. The fixed-node policy is temporary; the returned graph stores the resulting coordinates as `pos`, but it does not turn them into persistent `pin` constraints. This is useful after one node-placement pass when a later pass should route or relax selected edges without disturbing the node layout:

```
#let g = graph.parse("digraph partial { a [pos=\"0,0\"]; b [pos=\"4,0\"]; c [pos=\"8,0\"]; a ->
b; b -> c }").at(0)
#let first = subgraph.bits(g, (true, true, false, false))
#let g = layout(g, layout-algo: "tree", layout-nodes: "fixed", subgraph: first)
#draw(g)
```

The shared spring/charge model uses the following coefficients:

$$c_{\mathrm{vv}} = \beta L^2$$

$$c_{\mathrm{ev}} = \beta \gamma_{\mathrm{ev}} L^2$$

$$c_{\mathrm{ee}} = \beta \gamma_{\mathrm{ee}} L^2$$

$$c_{\mathrm{center}} = \beta g_{\mathrm{center}} L^2$$

$$c_{\mathrm{dangling}} = \beta \gamma_{\mathrm{dangling}} L^2$$

The Typst parameter names are `beta` for $\beta$, `gamma-ev` for $\gamma_{\mathrm{ev}}$, `gamma-ee` for $\gamma_{\mathrm{ee}}$, `g-center` for $g_{\mathrm{center}}$, and `gamma-dangling` for $\gamma_{\mathrm{dangling}}$. The spring stiffness $k$ is `k-spring`, and the softening constant $\varepsilon$ is `eps`.

In `layout-algo: "anneal"`, linnest minimizes an energy:

$$E = \sum_{i<j} \frac{1}{2} \frac{c_{\mathrm{vv}}}{d(v_i, v_j) + \varepsilon} + \sum_{i,e} \frac{c_{\mathrm{ev}}}{d(v_i, e) + \varepsilon} + \sum_{(v,e)\ \mathrm{incident}} \frac{1}{2} k (\ell_e - d(v,e))^2$$
$$+ \sum_{\mathrm{local\ edge\ pairs}} \frac{1}{2} \frac{c_{\mathrm{ee}}}{d(e_i, e_j) + \varepsilon} + \sum_{\mathrm{dangling\ pairs}} \frac{1}{2} \frac{c_{\mathrm{dangling}}}{d(e_i, e_j) + \varepsilon} + \sum_i \frac{1}{2} \frac{c_{\mathrm{center}}}{d(v_i, 0) + \varepsilon} + p_{\mathrm{cross}} N_{\mathrm{cross}}$$

.

Here $p_{\mathrm{cross}}$ is `crossing-penalty` and $N_{\mathrm{cross}}$ is the number of detected edge crossings. `temp`, `step`, `seed`, `steps`, `epochs`, `cool`, `accept-floor`, `step-shrink`, and `incremental-energy` belong to this simulated annealing mode. `crossing-penalty` is also anneal-only; force mode does not currently add a crossing force.

In `layout-algo: "force"`, linnest applies the direct forces corresponding to the same vertex-vertex, edge-vertex, incidence spring, local edge-edge, dangling-edge, and center terms. `step` is the integration step, `delta` clamps per-step movement, `steps` and `epochs` set the iteration budget, `cool` shrinks the step after

each epoch, and `early-tol` stops when movement is small. `z-spring` and `z-spring-growth` are force-only helpers: the integrator gives points temporary z coordinates to break overlaps and pulls them back toward the 2D plane.

`directional-force` is applied in both modes as an extra bias derived from pin/port direction constraints.

After either graph layout mode, labels are relaxed separately. If $L_l$ is the label target distance and $q_l$ is the label repulsion strength, then $L_l = \alpha_l L$ and $q_l = \beta_l L^2$. The Typst names are `label-length-scale` for $\alpha_l$ and `label-charge` for $\beta_l$. `label-spring` is the spring constant pulling each label toward its target. `label-steps`, `label-step`, `label-early-tol`, and `label-max-delta-scale` control the label relaxation iteration.

## Subgraphs

Subgraph objects are opaque zero-copy values.

- `subgraph.label(g, label)` constructs a subgraph from a base62 label.
- `subgraph.bits(g, bits)` constructs a subgraph from a boolean hedge array.
- `subgraph.compass(g, compass)` selects half edges with a DOT compass point.
- `subgraph.to-label(sg)` returns the base62 label.
- `subgraph.hedges(sg)` returns included hedge indices.
- `subgraph.contains(sg, hedge)` tests hedge membership.

## Generated Reference

## draw

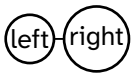
- draw()
- edge-halves()
- to-cetz-edge-halves()

### draw

Draw a graph object with CeTZ.

The graph must already have positions, either from `layout` or from explicit `pos` values passed to graph node or edge items. Paired edges without an explicit edge position use the midpoint of their endpoint nodes. Node and edge Typst style dictionaries or callbacks are evaluated and forwarded to CeTZ.

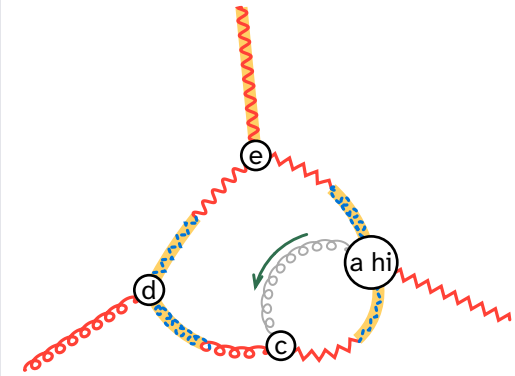
```
#let positioned = graph.build({
  graph.node(<left>, label: "left", pos: graph.pos(x: 0, y: 0))
  graph.node(<right>, label: "right", pos: graph.pos(ref: <left>, dx: 2.5, dy: 0))
  graph.edge(graph.source(<left>), graph.sink(<right>))
  graph.edge(graph.source(<right>), <right-out>, pos: graph.pos(ref: <right>, dx: 0.9, dy: 0.7))
},
  name: "positioned",
)
#draw(positioned, source-style: (stroke: black + 0.7pt), sink-style: (stroke: black + 0.7pt))
```




```
#let parallel-base-edge = 0
#let source-patterns = (
  "coil",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
  "coil",
  "wave",
  "zigzag",
)
```

```
#let sink-patterns = (
  "coil",
  "zigzag",
  "coil",
  "zigzag",
  "coil",
  "wave",
  "coil",
  "wave",
)
#let parallel-layer(edge, mark: none) = if edge.eid == parallel-base-edge {
  (
    offset: -0.5,
    length: 1.5,
    ratio: 0.5,
    resolve-length: "min",
    stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
    mark: mark,
  )
} else { none }
#let stack(base, layer) = if layer == none { base } else { (base, layer) }
#let source-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: red, thickness: 1.2pt, cap: "round")
}
#let sink-stroke(edge) = if edge.eid == parallel-base-edge {
  (paint: gray, thickness: 0.75pt, cap: "round")
} else {
  (paint: blue, thickness: 1.2pt, cap: "round", dash: "dotted")
}
#let source-style(edge) = {
  let base = (
    stroke: source-stroke(edge),
    pattern: source-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge))
}
#let sink-style(edge) = {
  let base = (
    stroke: sink-stroke(edge),
    pattern: sink-patterns.at(edge.eid),
    pattern-amplitude: 0.18,
    pattern-wavelength: 0.55,
    pattern-coil-longitudinal-scale: 1.6,
  )
  stack(base, parallel-layer(edge, mark: (end: (symbol: "straight"), scale: 0.75)))
}
#let g = graph.build({
  graph.node(<a>, label: "a hi")
  graph.node(<c>)
  graph.node(<d>)
  graph.node(<e>)
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), pos: graph.pos(x: 0, y: 0.75, mode: "pin"))
  graph.edge(graph.source(<c>), <ca>, graph.sink(<a>, compass: "e"))
  graph.edge(graph.source(<c>), <cd>, graph.sink(<d>, compass: "e"))
  graph.edge(graph.source(<e>), <ed>, graph.sink(<d>, compass: "e"))
  graph.edge(graph.source(<e>), <ea>, graph.sink(<a>, compass: "e"))
  graph.edge(graph.source(<d>), <d-out>)
  graph.edge(graph.source(<e>, compass: "e"), <e-out>)
  graph.edge(graph.source(<a>), <a-out>)
},
  name: "demo",
)
#let layed-out = layout(g)
#let east = subgraph.compass(layed-out, "e")
```

```
#draw(layed-out, subgraph: east, source-style: source-style, sink-style: sink-style)
```

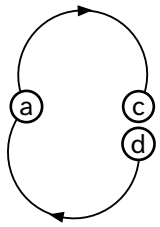



```
#let p = graph.build({
  graph.node(<pa>, label: "a", pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc>, label: "c", pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc1>, label: "c", pos: graph.pos(y: 0, mode: "pin"))
  graph.node(<pc2>, label: "c", pos: graph.pos(y: 0, mode: "pin"))
  graph.edge(graph.source(<pa>), <e0>, graph.sink(<pc>))
  graph.edge(graph.source(<pc2>), <e1>, graph.sink(<pc1>))
},
  name: "parallel demo",
)
#let parallel-edge-style(edge) = (
    offset: -0.5,
    length: 1.4,
    ratio: 0.5,
    resolve-length: "min",
    stroke: (paint: rgb("#2f6f4e"), thickness: 1.1pt, cap: "round"),
  )

#let focused-base-style(edge) = (
  stroke: (paint: gray, thickness: 0.7pt, cap: "round"),
  pattern: "coil",
  pattern-amplitude: 0.14,
  pattern-wavelength: 0.45,
  pattern-coil-longitudinal-scale: 1.5,
)
#let focused-source-style(edge) = (focused-base-style(edge), parallel-edge-style(edge))
#let focused-sink-style(edge) = (focused-base-style(edge), parallel-edge-style(edge) + (
  mark: (end: ">"),
))
#draw(layout(p, g-center: 0.004, label-steps: 0,), unit: 1.25, source-style: focused-source-style, sink-
style: focused-sink-style)
```


```
#let g = graph.build({
  graph.node(<a>, pos: graph.pos(x: 0, y: 0, mode: "pin"))
  graph.node(<c>, pos: graph.pos(x: 3, y: 0, mode: "pin"))
  graph.node(<d>, pos: graph.pos(x: 3, y: -1, mode: "pin"))
  graph.edge(graph.source(<a>), <ac>, graph.sink(<c>), orientation: "default")
  graph.edge(graph.source(<a>), <ad>, graph.sink(<d>), orientation: "reversed")
},
  name: "oriented marks",
```

```
)
#let arrow = (
  end: (symbol: ">", fill: black, anchor: "center", shorten-to: auto),
  scale: 0.75,
)
#let oriented-arrow = (
  stroke: black + 0.7pt,
  mark: arrow,
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
#draw(layout(g), unit: 1.4, source-style: oriented-arrow, sink-style: oriented-arrow)
```




### Parameters

```
draw(
  graph: bytes,
  scope: dictionary,
  unit: int float length ratio,
  title: none auto content string,
  subgraph: none bytes,
  debug: bool int,
  node-radius: auto int float array,
  node-min-radius: int float,
  node-label-padding: int float,
  node-fill: any,
  node-stroke: any,
  node-outset: auto int float,
  node-label-style: dictionary,
  node-style: dictionary function none,
  node-label: auto content string function none,
  edge-stroke: any,
  edge-offset: int float,
  edge-length: none int float,
  edge-ratio: none int float,
  edge-resolve-length: string function,
  edge-accuracy: float,
  edge-optimize: bool,
  source-style: dictionary array function none,
  sink-style: dictionary array function none,
  edge-label: content string function none,
  edge-omega: float,
  edge-trim-accuracy: float,
  padding: none int float array dictionary,
  debug-edge-radius: int float,
  debug-edge-fill: any,
  debug-edge-stroke: any,
  debug-edge-label-fill: any,
  subgraph-edge-style: dictionary,
  subgraph-edge-underlay: bool
) -> content
```

**graph** `bytes`

Graph object with positions from `layout` or explicit graph API `pos` fields.

**scope** `dictionary`

Additional values merged into node and edge callback dictionaries.

Default: `(:)`

**unit** `int` or `float` or `length` or `ratio`

Coordinate length for one graph-layout unit. Numbers are interpreted as em.

Default: `1`

**title** `none` or `auto` or `content` or `string`

Optional title displayed above the diagram. Use `auto` for the graph name.

Default: `none`

**subgraph** `none` or `bytes`

Optional subgraph whose half-edges are shaded.

Default: `none`

**debug** `bool` or `int`

Debug level. `1` enables CeTZ canvas debug; `2` also marks edge positions.

Default: `false`

**node-radius** `auto` or `int` or `float` or `array`

Default CeTZ node radius. Use `auto` to fit the node label.

Default: `auto`

**node-min-radius** `int` or `float`

Minimum radius used when `node-radius` is `auto`.

Default: `0.16`

**node-label-padding** `int` or `float`

Extra canvas-unit padding around labels when `node-radius` is `auto`.

Default: `0.08`

**node-fill** `any`

Default CeTZ node fill.

Default: `white`

**node-stroke** `any`

Default CeTZ node stroke.

Default: `black`

**node-outset** `auto` or `int` or `float`

Edge clearance from node centers. `auto` uses each node circle radius. Increase this when labels extend beyond the circle.

Default: `auto`

**node-label-style** `dictionary`

Default node-label style forwarded to `cetz.draw.content`.

Default: `(:)`

**node-style** `dictionary` or `function` or `none`

Node style dictionary or callback. A callback receives node data.

Default: `(:)`

**node-label** `auto` or `content` or `string` or `function` or `none`

Node label content or callback. `auto` uses the node name.

Default: `auto`

**edge-stroke** `any`

Default CeTZ edge stroke.

Default: `0.1em`

**edge-offset** `int` or `float`

Default normal offset for edge paths. Applied to the base edge geometry before patterns; node outsets then trim the shifted path.

Default: `_edge-geometry-defaults.offset`

**edge-length** `none` or `int` or `float`

Maximum visible arc length for centered parallel edge paths. `none` keeps the full shifted path.

Default: `_edge-geometry-defaults.length`

**edge-ratio** `none` or `int` or `float`

Maximum visible fraction of the base edge length for centered parallel edge paths. Combined with `edge-length` according to `edge-resolve-length`.

Default: `_edge-geometry-defaults.ratio`

**edge-resolve-length** `string` or `function`

Resolve `edge-length` and `edge-ratio`. Accepted string values are `"min"`/`"shorter"`, `"max"`/`"longer"`, `"length"`/`"fixed"`, `"ratio"`/`"relative"`, or `"none"`/`"full"`. A function receives `(base-length, length, ratio)`.

Default: `_edge-geometry-defaults.resolve-length`

**edge-accuracy** `float`

Arc-length accuracy for fitted parallel edge paths.

Default: `_edge-geometry-defaults.accuracy`

**edge-optimize** `bool`

Let Kurbo optimize the fitted parallel path.

Default: `_edge-geometry-defaults.optimize`

**source-style** `dictionary` or `array` or `function` or `none`

Source half-edge style dictionary, array of layer dictionaries, or callback. `mark-position: "center-if-dangling"` keeps an end marker at the paired-edge split point while centering it on dangling half edges. `mark-orientation: "edge"` makes a mark follow `edge.orientation` instead of raw path direction; reversed edges move the mark to the sink half and flip it, while undirected edges suppress it.

Default: `(:)`

**sink-style** `dictionary` or `array` or `function` or `none`

Sink half-edge style dictionary, array of layer dictionaries, or callback. A callback receives edge data. `mark-position: "center-if-dangling"` has the same dangling-edge behavior as for `source-style`, and `mark-orientation: "edge"` participates in the same orientation-aware mark placement.

Default: `(:)`

**edge-label** `content` or `string` or `function` or `none`

Edge label content or callback. A callback receives edge data.

Default: `none`

**edge-omega** `float`

Hobby curl used at the endpoints of paired edge curves.

Default: `1.0`

**edge-trim-accuracy** `float`

Arc-length accuracy for trimming edge curves at node outsets.

Default: `0.001`

**padding** `none` or `int` or `float` or `array` or `dictionary`

CeTZ canvas padding.

Default: `0.4`

**debug-edge-radius** `int` or `float`

Radius for edge-position markers shown at `debug >= 2`.

Default: `0.08`

**debug-edge-fill** `any`

Fill for edge-position markers shown at `debug >= 2`.

Default: `rgb("#ff9f1c")`

**debug-edge-stroke** `any`

Stroke for edge-position markers shown at `debug >= 2`.

Default: `rgb("#d72638") + 0.35pt`

**debug-edge-label-fill** `any`

Label fill for edge-position markers shown at `debug >= 2`.

Default: `rgb("#7a1020")`

**subgraph-edge-style** `dictionary`

CeTZ style used to shade half-edges included in `subgraph`.

Default: `(stroke: rgb("#ffd166") + 4.5pt)`

**subgraph-edge-underlay** `bool`

Draw subgraph shading below the normal half-edge style.

Default: `true`

**edge-halves**

Split a laid-out graph edge into source and sink half-edge paths.

The returned dictionary has `source`, `sink`, and `curve`. The split point is the edge layout point, so the two half-edges join smoothly there.

**Parameters**

```
edge-halves(
  edge,
  nodes,
  omega,
  source-outset,
  sink-outset,
  accuracy
) -> dictionary
```

**to-cetz-edge-halves**

Draw the two halves of a laid-out graph edge through CeTZ.

`source-style` applies from the source node to the edge layout point, and `sink-style` applies from the edge layout point to the sink node.

**Parameters**

```
to-cetz-edge-halves(
  edge,
  nodes,
  unit,
  omega,
  source-outset,
  sink-outset,
  accuracy,
  source-style,
  sink-style
) -> content
```

- layout()

**Variables**

- graph
- node
- edge
- source
- sink
- curve
- physics
- subgraph

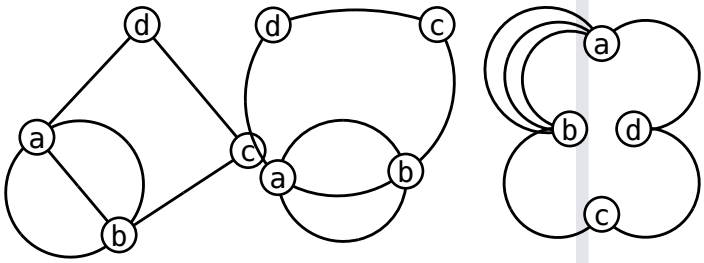
**layout**

Apply the linnest layout pass to a graph object.

This is intentionally a second step: construct or parse a graph first, then call `layout`. Set `layout-algo` to `"force"` for deterministic force integration, `"anneal"` for simulated annealing, `"tree"` for a traversal tree placement, or `"dot"` for a layered directed placement.

```
#let g = graph.parse("digraph partial { a:s ->
b:s;a -> b;a -> b; b:s -> c:s; c:s -> d:s; d:s -
> a:s }").at(0)
#let south = subgraph.compass(g,"s")
#let gf = layout(layout(g, layout-algo:
"force"), layout-algo: "tree", layout-nodes:
"fixed",subgraph:south)
#let ga = layout(g, layout-algo: "anneal")
#let gt = layout(layout(g, layout-algo:
"tree",layout-roots: (2)), layout-algo:
```

```
"anneal", layout-nodes: "fixed",gamma-
ee:0.1,gamma-ev:1.5,beta:20,length-scale:0.4)
#grid(columns: 3, gutter: 2cm, draw(gf),
draw(ga), draw(gt))
```




```
#let g = graph.parse("digraph partial { a -> b;
b -> c; c -> d; d -> a }").at(0)
#let tree = graph.forests(g).at(0)
#let g = layout(g, layout-algo: "tree",
subgraph: tree)
#graph.edges(g).map(edge => edge.pos)
```

```
(
  (x: 0.0, y: 1.5),
  (x: 0.0, y: 4.5),
  (x: 0.0, y: 7.5),
  (x: 0.0, y: 4.5),
)
```

### Parameters

```
layout(
  graph: bytes,
  subgraph: none bytes,
  viewport-w: float,
  viewport-h: float,
  tree-dx: float,
  tree-dy: float,
  steps: int,
  seed: int,
  step: float,
  step-shrink: float,
  cool: float,
  accept-floor: float,
  early-tol: float,
  temp: float,
  delta: float,
  beta: float,
  k-spring: float,
  g-center: float,
  epochs: int,
  crossing-penalty: float,
  gamma-dangling: float,
  gamma-ee: float,
  directional-force: float,
  label-length-scale: float,
  label-spring: float,
  label-charge: float,
  label-steps: int,
  label-step: float,
  label-early-tol: float,
  label-max-delta-scale: float,
  gamma-ev: float,
  eps: float,
  incremental-energy: bool,
  layout-algo: string,
  layout-nodes: string,
  layout-roots: array,
  z-spring: float,
  z-spring-growth: float,
  length-scale: float
) -> bytes
```

**graph** `bytes`

Graph object returned by `graph.build` or `graph.parse`.

**subgraph** `none` or `bytes`

Optional subgraph object to lay out. With `"tree"` and `"dot"`, other edges are drawn from the resulting node positions as straight lines. With `"force"` and `"anneal"`, nodes and edges outside the subgraph are fixed boundary points during optimization.

Default: `none`

**viewport-w** `float`

Width of the layout viewport used to derive the natural spring length. Applies to both `"force"` and `"anneal"`.

Default: `10.0`

**viewport-h** `float`

Height of the layout viewport used to derive the natural spring length. Applies to both `"force"` and `"anneal"`.

Default: `10.0`

**tree-dx** `float`

Horizontal spacing multiplier for the initial traversal-tree placement. Applies before both `"force"` and `"anneal"`.

Default: `0.9`

**tree-dy** `float`

Vertical spacing multiplier for the initial traversal-tree placement. Applies before both `"force"` and `"anneal"`.

Default: `1.2`

**steps** `int`

Iterations per epoch. In `"force"` mode this is the number of force integration steps; in `"anneal"` mode this is the number of proposals per temperature epoch.

Default: `int(sys.inputs.at("steps", default: "30"))`

**seed** `int`

Seed for deterministic initialization, force-mode jitter, and annealing proposals. Applies to both modes.

Default: `int(sys.inputs.at("seed", default: "2"))`

**step** `float`

Initial movement scale. `"force"` multiplies computed forces by this value; `"anneal"` uses it as the proposal step size.

Default: `0.81`

**step-shrink** `float`

Anneal-only step shrink factor, applied when an epoch's acceptance ratio falls below `accept-floor`.

Default: `0.21`

**cool** `float`

Cooling factor applied once per epoch. `"anneal"` cools `temp`; `"force"` shrinks `step`.

Default: `0.85`

**accept-floor** `float`

Anneal-only acceptance-ratio threshold below which `step` is shrunk by `step-shrink`.

Default: `0.15`

**early-tol** `float`

Force-only early stop threshold for maximum movement in one step. The annealing schedule stores this value but does not currently use it for stopping.

Default: `1e-6`

**temp** `float`

Anneal-only initial temperature used in the Metropolis acceptance test.

Default: `0.3`

**delta** `float`

Force-mode maximum movement clamp per point and per step.

Default: `0.4`

**beta** `float`

Base repulsion strength for vertex-vertex interactions. Also scales `gamma-ev`, `gamma-ee`, `gamma-dangling`, and `g-center`. Applies to both modes through the shared spring energy.

Default: `50.0`

**k-spring** `float`

Spring stiffness for node-to-edge incidence lengths. Applies to both modes.

Default: `11.0`

**g-center** `float`

Centering strength relative to `beta`. Applies to both modes.

Default: `0.005`

**epochs** `int`

Number of epochs. Both modes run up to `steps` iterations inside each epoch.

Default: `30`

**crossing-penalty** `float`

Anneal-only fixed energy penalty per detected edge crossing. The direct force integrator does not currently add a crossing force.

Default: `30.0`

**gamma-dangling** `float`

Repulsion for dangling half edges, relative to `beta`. Applies to both modes through the shared spring energy.

Default: `5.0`

**gamma-ee** `float`

Local edge-edge repulsion, relative to `beta`. Applies to both modes.

Default: `0.1`

**directional-force** `float`

Bias that pushes points in directions implied by pin/port constraints. Applies to both modes.

Default: `5.0`

**label-length-scale** `float`

Edge-label target offset as a multiple of the graph spring length. Label layout runs after both graph layout modes.

Default: `0.6`

**label-spring** `float`

Spring strength pulling each label toward its target offset. Label layout runs after both modes.

Default: `23.0`

**label-charge** `float`

Repulsion strength between labels and graph points, scaled by spring length squared. Label layout runs after both modes.

Default: `3.0`

**label-steps** `int`

Maximum number of post-layout label relaxation steps. Set to `0` to skip label placement. Applies after both modes.

Default: `20`

**label-step** `float`

Label relaxation step size. Applies after both modes.

Default: `0.15`

**label-early-tol** `float`

Label relaxation early stop threshold. Applies after both modes.

Default: `1e-3`

**label-max-delta-scale** `float`

Label movement clamp as a multiple of spring length. Applies after both modes.

Default: `0.5`

**gamma-ev** `float`

Edge-vertex repulsion, relative to `beta`. Applies to both modes.

Default: `0.01`

**eps** `float`

Softening epsilon used in inverse-square force/energy terms. Applies to both modes.

Default: `1e-4`

**incremental-energy** `bool`

Whether annealing updates cached energy by local deltas. This is anneal-only; force mode computes direct forces instead of energies.

Default: `true`

**layout-algo** `string`

Layout algorithm. Use `"force"` for direct force integration, `"anneal"` for simulated annealing against the spring energy, `"tree"` for a traversal-tree placement, or `"dot"` for a layered directed placement.

Default: `"force"`

**layout-nodes** `string`

Node movement policy. `"layout"` lets the layout algorithm move nodes. `"fixed"` keeps every node at its current position for this layout pass and only moves edge control points. With `subgraph`, only edges in the subgraph are moved; other edge control points stay at their current positions.

Default: `"layout"`

**layout-roots** `array`

Ordered node indices used as preferred roots for `"tree"` and `"dot"`. Roots outside the selected node set are ignored. Remaining components are laid out afterward in graph order.

Default: `()`

**z-spring** `float`

Force-only spring pulling temporary z coordinates back toward the layout plane. Use with `z-spring-growth` to help separate overlapping points during integration.

Default: `0.05`

**z-spring-growth** `float`

Force-only per-epoch multiplier for `z-spring`.

Default: `1.3`

**length-scale** `float`

Natural spring-length multiplier. This scales the graph's preferred edge length and the repulsive coefficients derived from it. Applies to both modes.

Default: `0.5`

**graph** `module`

Graph namespace.

Graph objects are opaque zero-copy values. Keep them opaque and pass them back to this module, `subgraph`, or `layout()`.

**node** `array`

Create a graph node item for `graph.build`.

**edge** `array`

Create a graph edge item for `graph.build`.

**source** `dictionary`

Create a source half-edge endpoint for `edge`.

**sink** `dictionary`

Create a sink half-edge endpoint for `edge`.

### curve `module`

Curve namespace.

Helpers for splitting and drawing Bezier edge geometry.

### physics `module`

Physics edge-style namespace.

Reusable particle-line styles and callbacks for diagrams drawn with `draw`.

### subgraph `module`

Subgraph namespace.

Subgraph objects are opaque zero-copy values. They can be passed to `graph.nodes` and `graph.edges`.

## graph

- build()
- cycles()
- dot()
- edge()
- edges()
- forests()
- group()
- info()
- join()
- node()
- nodes()
- parse()
- pos()
- sink()
- source()

**build**

Build one graph object from a stream of node and edge items.

Use `node()`, `source()`, `sink()`, and `edge()` to create graph items. Positional items may be passed as comma-separated arguments or yielded from a Typst code block.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), <e>,
graph.sink(<b>))
}, name: "demo")
#graph.info(g).name
```

```
demo
```

### Parameters

```
build(
  ..items: array,
  name: none string,
  statements: dictionary,
  edge-statements: dictionary,
  source-style-eval: none string,
  sink-style-eval: none string,
  label-eval: none string,
  node-statements: dictionary,
  nodes: array,
  edges: array
) -> bytes
```

#### ..items `array`

Node and edge items returned by `node()` and `edge()`.

#### name `none` or `string`

Graph name.

Default: `none`

#### statements `dictionary`

Graph-level DOT statements.

Default: `(:)`

#### edge-statements `dictionary`

Default DOT statements for edges. String values may use `{name}` placeholders that expand from each edge's merged statement dictionary.

Default: `(:)`

#### source-style-eval `none` or `string`

Default always-evaluated Typst style for source half-edges. This is merged into `edge-statements` as `source-style-eval`.

Default: `none`

#### sink-style-eval `none` or `string`

Default always-evaluated Typst style for sink half-edges. This is merged into `edge-statements` as `sink-style-eval`.

Default: `none`

**label-eval** `none` or `string`

Default always-evaluated Typst edge label template. This is merged into `edge-statements` as `label-eval`.

Default: `none`

**node-statements** `dictionary`

Default DOT statements for nodes.

Default: `(:)`

**nodes** `array`

Additional node items or raw node specs. -> array

Default: `()`

**edges** `array`

Additional edge items or raw edge specs.

Default: `()`

### cycles

Return subgraph objects for the graph's cycle basis.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#graph.cycles(g).len()
```

```
0
```

**Parameters**

`cycles(graph) -> array`

### dot

Serialize a graph object to DOT.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
}, name: "demo")
#graph.dot(g).contains("digraph demo")
```

```
true
```

**Parameters**

`dot(graph) -> string`

### edge

Create a graph edge item for `build()`.

Positional arguments may contain one `source()`, one `sink()`, and optionally one Typst label used as the edge name. The numeric `id` chooses the edge order and the visible/DOT label is `label`.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), <e>,
graph.sink(<b>), label: "p")
})
```



#### Parameters

```
edge(
  ..args: any,
  name: none label,
  id: none int,
  orientation: string,
  label: none string,
  pos: none dictionary,
  shift: none string array dictionary,
  label-pos: none string array dictionary,
  label-angle: none int float string,
  bend: none int float string,
  statements: dictionary,
  source-style-eval: none string,
  sink-style-eval: none string,
  label-eval: none string
) -> array
```

**..args** `any`

Source/sink half-edges and optional edge name.

**name** `none` or `label`

Typst edge name.

Default: `none`

**id** `none` or `int`

Numeric edge order/index override.

Default: `none`

**orientation** `string`

Edge orientation: `"default"`, `"reversed"`, or `"undirected"`.

Default: `"default"`

**label** `none` or `string`

Visible/DOT edge label, stored in `statements.label`.

Default: `none`

**pos** `none` or `dictionary`

Edge placement.

Default: `none`

**shift** `none` or `string` or `array` or `dictionary`

Drawing shift stored as a statement.

Default: `none`

**label-pos** `none` or `string` or `array` or `dictionary`

Edge label position stored as a statement.

Default: `none`

**label-angle** `none` or `int` or `float` or `string`

Edge label angle stored as a statement.

Default: `none`

**bend** `none` or `int` or `float` or `string`

Edge bend stored as a statement.

Default: `none`

**statements** `dictionary`

Additional DOT edge statements.

Default: `(:)`

**source-style-eval** `none` or `string`

Always-evaluated Typst style for the source half-edge.

Default: `none`

**sink-style-eval** `none` or `string`

Always-evaluated Typst style for the sink half-edge.

Default: `none`

**label-eval** `none` or `string`

Always-evaluated Typst edge label template.

Default: `none`

### edges

Return edge records, optionally filtered by an subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"),
graph.sink(<b>))
})
#let east = subgraph.compass(g, "e")
#graph.edges(g, subgraph: east).len()
```

```
1
```

**Parameters**

```
edges(
  graph,
  subgraph
) -> array
```

### forests

Return subgraph objects for the graph's spanning forests.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#graph.forests(g).len()
```

```
1
```

**Parameters**

```
forests(graph) -> array
```

### group

Create a grouped placement coordinate.

`side: "+"` keeps the solved coordinate non-negative and `side: "-"` keeps it non-positive. Groups are layout constraints and therefore require pin placement, which is the `graph.pos` default.

```
#graph.group("right", side: "+")
```

```
(kind: "group", name: "right", side: "+")
```

**Parameters**

```
group(
  name,
  side
) -> dictionary
```

### info

Return graph metadata.

The result has `name`, `global-statements`, `edge-statements`, and `node-statements`.

```
#let g = graph.build({ graph.node(<a>) }, name:
"demo")
```

```
#graph.info(g).name
```

```
demo
```

**Parameters**

```
info(graph) -> dictionary
```

### join

Join two graphs by matching dangling half-edge data on `key`.

Supported key values are `"statement"`, `"compass"`, and `"id"`.

```
#let left = graph.build({
  graph.node(<a>)
  graph.edge(graph.sink(<a>, statement: "j"))
})
#let right = graph.build({
  graph.node(<b>)
  graph.edge(graph.source(<b>, statement: "j"))
})
#graph.edges(graph.join(left, right, key:
"statement")).len()
```

```
1
```

**Parameters**

```
join(
  left,
  right,
  key
) -> bytes
```

### node

Create a graph node item for `build()`.

A Typst label is the node name used by `source()`, `sink()`, and `pos()`. The optional numeric `id` chooses the node order/index. `label` is display metadata stored in the DOT `label` statement.

```
#let g = graph.build({
  graph.node(<a>, id: 0, label: "A")
})
#graph.nodes(g).first().name
```

```
a
```

**Parameters**

```
node(
  ..args,
  name: none label,
  id: none int,
  label: none string,
  pos: none dictionary,
  shift: none string array dictionary,
  statements: dictionary
) -> array
```

**name** `none` or `label`

Typst node name for references.

Default: `none`

**id** `none` or `int`

Numeric node order/index override.

Default: `none`

**label** `none` or `string`

Visible/DOT node label, stored in `statements.label`.

Default: `none`

**pos** `none` or `dictionary`

Node placement.

Default: `none`

**shift** `none` or `string` or `array` or `dictionary`

Drawing shift stored as a statement.

Default: `none`

**statements** `dictionary`

Additional DOT node statements.

Default: `(:)`

### nodes

Return node records, optionally filtered by an subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
})
#graph.nodes(g).map(node => node.name).join(",
")
```

```
a, b
```

**Parameters**

```
nodes(
  graph,
  subgraph
) -> array
```

### parse

Parse one or more DOT digraphs into graph objects.

```
#let graphs = graph.parse("digraph first { a -> b }")
#graphs.len()
```

```
1
```

#### Parameters

```
parse(input) -> array
```

### pos

Create a first-class graph placement.

The default `mode: "pin"` turns numeric and grouped coordinates into layout constraints and also makes the coordinates immediately drawable without a layout pass. Use `mode: "start"` when the coordinate should only seed the layout. `ref` may reference a previously created node index and combines with `dx` and `dy`.

```
#graph.pos(x: 0, y: graph.group("row"), mode: "pin")
```

```
(
  mode: "pin",
  x: 0,
  y: (kind: "group", name: "row", side: none),
)
```

#### Parameters

```
pos(
  x,
  y,
  ref,
  dx,
  dy,
  mode
) -> dictionary
```

### sink

Create a sink half-edge endpoint.

`node` may be a node name like `<a>` or a numeric node index. `name` gives the half-edge a Typst name; `id` is a numeric half-edge order/index override.

```
#graph.sink(<b>, name: <h2>, id: 1, compass: "w")
```

```
(
  linnest-kind: "sink",
  node: <b>,
  name: <h2>,
  id: 1,
  statement: none,
  compass: "w",
)
```

**Parameters**

```
sink(
  node,
  name,
  id,
  statement,
  compass
) -> dictionary
```

**source**

Create a source half-edge endpoint.

`node` may be a node name like <a> or a numeric node index. `name` gives the half-edge a Typst name; `id` is a numeric half-edge order/index override.

```
#graph.source(<a>, name: <h1>, id: 0, compass:
"e")
```

```
(
  linnest-kind: "source",
  node: <a>,
  name: <h1>,
  id: 0,
  statement: none,
  compass: "e",
)
```

**Parameters**

```
source(
  node,
  name,
  id,
  statement,
  compass
) -> dictionary
```

## physics

- dangling-half-edge-index()
- edge-entry()
- edge-index()
- edge-label()
- eval-template()
- interpolate-template()
- label-content()
- momentum-value()
- particle-name()
- sink-stroke()
- sink-style()
- source-stroke()
- source-style()
- stroke-style()
- style()
- style-dict()
- text-value()

**Variables**

- massive
- massless

- dashed
- dotted
- fermion-arrow-mark
- fermion-flow
- wave
- coil
- zigzag
- default-edge
- default-map
- momentum-arrow-defaults

**dangling-half-edge-index**
Return the exposed half-edge index for dangling edges only.

**Parameters**
```
dangling-half-edge-index(edge) -> none any
```

**edge-entry**
Return the style-map entry for an edge.

**Parameters**
```
edge-entry(
  edge,
  map,
  default
) -> dictionary
```

**edge-index**
Return the edge index used by optional edge labels. The DOT `id` statement wins over the renderer-local `eid`.

**Parameters**
```
edge-index(
  edge,
  fields
) -> none any
```

**edge-label**
Edge-label callback.

By default this preserves explicit `label-eval`, `display-label`, `label-template`, `label`, and particle-map labels. Set any `show-*` option to build a label from selected metadata instead:

```
#let callbacks = physics.style(
  momentum-arrows: true,
  show-edge-index: true,
  show-particle: true,
)
```

**Parameters**

```
edge-label(
  edge,
  map,
  default,
  typst-fields,
  scope,
  show-momentum,
  show-edge-index,
  show-half-edge-index,
  show-particle,
  momentum-fields,
  edge-index-fields,
  momentum-prefix,
  edge-index-prefix,
  half-edge-index-prefix,
  particle-prefix,
  label-separator,
  ],
  label-size,
  label-fill
) -> none content
```

**eval-template**

Evaluate a string-valued style template after placeholder interpolation.

**Parameters**

```
eval-template(
  template,
  edge,
  map,
  scope
) -> any
```

**interpolate-template**

Replace `{field}` placeholders in a string-valued style template. `{{` and `}}` produce literal braces.

**Parameters**

```
interpolate-template(
  template,
  edge
) -> none string
```

**label-content**

Convert a style label value to content. In `mode: "eval"`, string values are evaluated after placeholder interpolation.

**Parameters**

```
label-content(
  value,
  edge,
  mode,
  map,
  scope
) -> none content
```

**momentum-value**

Return the momentum field used by optional edge labels.

**Parameters**

```
momentum-value(
  edge,
  fields
) -> none any
```

**particle-name**

Return an edge's particle name, stripping the quotes often present in DOT statement values.

**Parameters**

```
particle-name(edge) -> none string
```

**sink-stroke**

Sink-half stroke helper. The default lightening makes the two halves encode the graph's source/sink split without requiring arrowheads.

**Parameters**

```
sink-stroke(
  c,
  thickness,
  dash,
  lighten
) -> dictionary
```

**sink-style**

Style callback for the sink half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration toward the sink node, so momentum arrows always flow from source to sink independently of `edge.orientation`.

**Parameters**

```
sink-style(
  edge,
  map,
  default,
  typst-fields,
  scope,
  orientation-split,
  momentum-arrows,
  momentum-arrow-offset,
  momentum-arrow-length,
  momentum-arrow-ratio,
  momentum-arrow-stroke,
  momentum-arrow-mark
) -> dictionary array
```

**source-stroke**

Source-half stroke helper.

**Parameters**

```
source-stroke(
  c,
  thickness,
  dash
) -> dictionary
```

**source-style**

Style callback for the source half edge.

`momentum-arrows: true` adds one centered black CeTZ-mark decoration while the main edge remains drawn with its normal particle style.

**Parameters**

```
source-style(
  edge,
  map,
  default,
  typst-fields,
  scope,
  orientation-split,
  momentum-arrows,
  momentum-arrow-offset,
  momentum-arrow-length,
  momentum-arrow-ratio,
  momentum-arrow-stroke,
  momentum-arrow-mark
) -> dictionary array
```

**stroke-style**

Build a rounded CeTZ stroke dictionary accepted by `linnest.draw`.

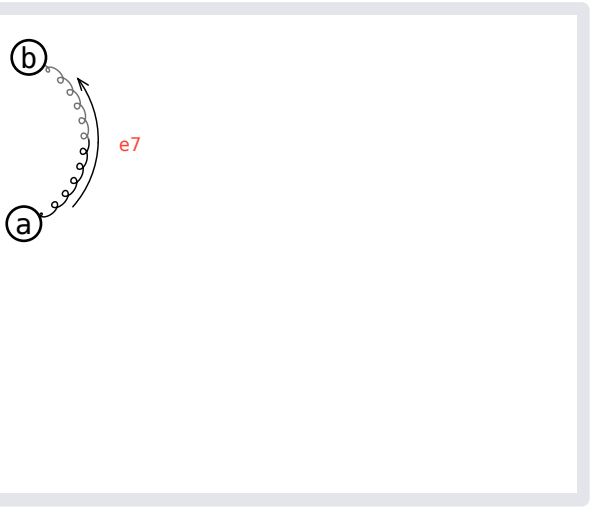
### Parameters

```
stroke-style(
  c,
  thickness,
  dash
) -> dictionary
```

### style

Bundle source-style, sink-style, and edge-label callbacks with shared options for `linnest.draw`.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), <g-edge>,
graph.sink(<b>), statements: (particle: "g", id:
"7"))
},
  name: "physics",
)
#let callbacks = physics.style(momentum-arrows:
true, show-edge-index: true)
#draw(
  layout(g),
  source-style: callbacks.source-style,
  sink-style: callbacks.sink-style,
  edge-label: callbacks.edge-label,
)
```




### Parameters

```
style(
  map,
  default,
  typst-fields,
  scope,
  orientation-split,
  momentum-arrows,
  momentum-arrow-offset,
  momentum-arrow-length,
  momentum-arrow-ratio,
  momentum-arrow-stroke,
  momentum-arrow-mark,
  show-momentum,
  show-edge-index,
  show-half-edge-index,
  show-particle,
  momentum-fields,
  edge-index-fields,
  momentum-prefix,
  edge-index-prefix,
  half-edge-index-prefix,
  particle-prefix,
  label-separator,
  ],
  label-size,
  label-fill
) -> dictionary
```

### style-dict

Convert a style value to a dictionary. In `mode: "eval"`, string values are evaluated after placeholder interpolation.

**Parameters**

```
style-dict(
  value,
  edge,
  mode,
  map,
  scope
) -> dictionary
```

### text-value

Convert DOT-ish values to plain text.

**Parameters**

```
text-value(value) -> string
```

### massive `length`

Conventional massive-particle stroke thickness.

### massless `length`

Conventional massless-particle stroke thickness.

### dashed `array`

Dash pattern used for scalar-style lines.

### dotted `string`

Dotted dash style used for ghost-style lines.

### fermion-arrow-mark `dictionary`

CeTZ marker used for fermion particle-flow arrows on the main edge.

### fermion-flow `dictionary`

Mark an edge-map entry as a fermion so the main edge receives one particle-flow arrow that follows `edge.orientation`. This is separate from optional momentum arrows.

**wave** `dictionary`

Photon-style wave pattern.

**coil** `dictionary`

Gluon-style coil pattern.

**zigzag** `dictionary`

Weak-boson-style zigzag pattern.

**default-edge** `dictionary`

Fallback style entry used when an edge has no known particle type.

**default-map** `dictionary`

Small built-in particle map for standalone physics diagrams. GammaLoop generated `edge-style.typ` files pass their model-specific map to `style`.

**momentum-arrow-defaults** `dictionary`

Default style for source-to-sink momentum arrow layers.

## subgraph

- bits()
- compass()
- contains()
- hedges()
- label()
- to-label()

### bits

Construct an subgraph object from a boolean hedge array.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#let first = subgraph.bits(g, (true, false))
#subgraph.contains(first, 0)
```

```
true
```

**Parameters**

```
bits(
  graph,
  bits
) -> bytes
```

### compass

Construct an subgraph object from a DOT compass point such as `"n"` or `"s"`.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"),
graph.sink(<b>))
})
#subgraph.hedges(subgraph.compass(g, "e")).len()
```

```
1
```

#### Parameters

```
compass(
  graph,
  compass
) -> bytes
```

### contains

Test whether an subgraph object includes a hedge.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.contains(subgraph.bits(g, (true,
false)), 0)
```

```
true
```

#### Parameters

```
contains(
  subgraph,
  hedge
) -> bool
```

### hedges

Return the hedge indices included in an subgraph object.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.hedges(subgraph.bits(g, (true,
false)))
```

```
(0,)
```

#### Parameters

```
hedges(subgraph) -> array
```

### label

Construct an subgraph object from a base62 label.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>, compass: "e"),
graph.sink(<b>))
})
#let east = subgraph.compass(g, "e")
#let same = subgraph.label(g, subgraph.to-
label(east))
#subgraph.hedges(same).len()
```

```
1
```

**Parameters**

```
label(
  graph,
  label
) -> bytes
```

### to-label

Convert an subgraph object to its base62 label.

```
#let g = graph.build({
  graph.node(<a>)
  graph.node(<b>)
  graph.edge(graph.source(<a>), graph.sink(<b>))
})
#subgraph.to-label(subgraph.bits(g, (true,
false)))
```

```
1
```

**Parameters**

```
to-label(subgraph) -> string
```